

Fig. 17.9. Gargantua and its disk, partially eclipsed by Miller's planet, as the Ranger, in the foreground, descends toward landing. *[From* Interstellar*, used courtesy of Warner Bros. Entertainment Inc.]*

Although I'm a scientist and aspire to science accuracy in science fiction, I can't blame Chris at all. I would have done the same, had I been making the decision. And you'd have thanked me for it.

---

31 The centrifugal force depends on the planet's orbital angular momentum, a measure of its orbital speed that is constant along its orbit (Chapter 10). In plotting how the force changes with distance from Gargantua in Figure 17.2, I hold that angular momentum constant. If the angular momentum were a bit smaller than the amount Miller's planet actually has, then the centrifugal force would everywhere be smaller, and the two curves in Figure 17.2 would not cross. There would be no balance point, and the planet would fall into Gargantua. That's why the location of Miller's planet in Figures 17.1 and 17.2 is the closest to Gargantua that the